Foundations: The Transformer That Kimi Builds On

Deep Dive into Kimi K3 — Chapter 1

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Foundation Books Project

The one idea of this chapter

A decoder turns an accepted prefix into a next-token prediction,
then uses prediction errors to change the model
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Goal: trace one fully declared four-token decoder from shifted targets to logits, through a cache, to a loss, gradients, and one AdamW update — every number checkable by hand and in the companion code.

Roadmap

The prediction task

Normalization and residuals

Causal attention and position

Channel mixing and the loss

Two executions of one prefix

From loss to one update

Toward Kimi K3

The prediction task

Shifted targets, one block, nine matrices

t	input	target
0	`BOS`	`red`
1	`red`	`square`
2	`square`	`EOS`

$T = 3$, $d = 4$, $H = 2$, $d_h = 2$, $m = 4$.

Identity-based weights, $W_O = \frac{1}{2} I$, $W_d = \frac{1}{4} I$: **144** trainable scalars.

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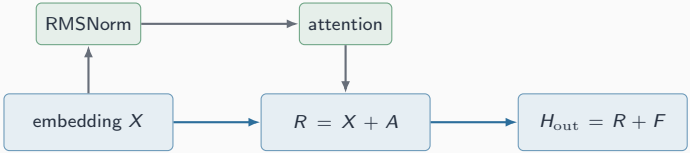
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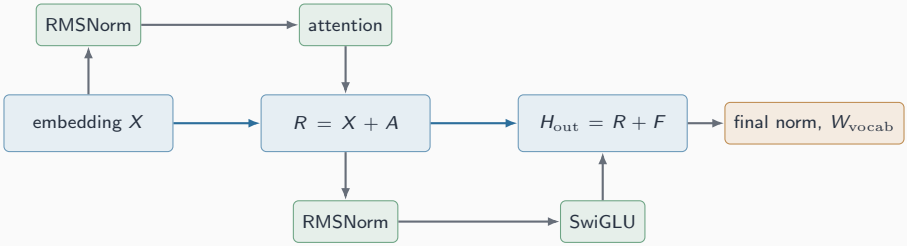
The target is the next token, shifted exactly once. Every weight is declared, so the chapter freezes the arithmetic — not a trained model.

Normalization and residuals

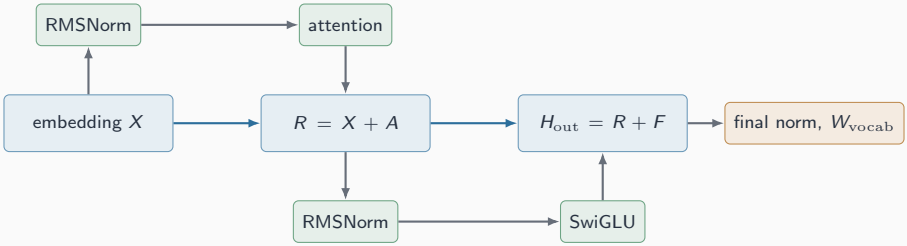
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RMSNorm reduces over one token's features; the residual adds the *original* input, not its normalized view. Pre-norm and post-norm are different functions.

Causal attention and position

Permitted keys, weighted values, rotated matching

$$e_{ts} = \frac{\langle q_t, k_s \rangle}{\sqrt{d_h}} + M_{ts}, \qquad \bar{q}_t^{\mathsf{T}} \bar{k}_s = q_t^{\mathsf{T}} \mathcal{R}\big((s-t)\theta\big) k_s$$

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Worked rows: $A_0 = (1, 0, 0, 0)$, $A_1 = (a, 1-a, 0, 0)$, $A_2 = (\frac{1}{3}, \frac{1}{3}, b, 0)$ — a nonzero weight can still carry a zero value.

Channel mixing and the loss

SwiGLU, two softmaxes, one mean loss

$$\mathrm{FFN}(u) = \big(\mathrm{SiLU}(uW_g) \odot uW_u\big) W_d, \qquad \mathcal{L} = \tfrac{1}{3}\textstyle\sum_t -\log p_t(y_t) \simeq 2.35161595$$

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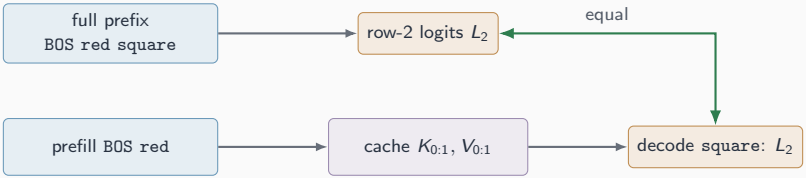
The untrained model prefers `BOS` at row 0 with 71% confidence: high confidence is not correctness.

Two executions of one prefix

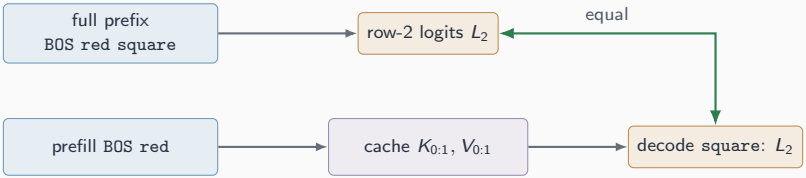
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Holds under the same weights, mask, and absolute positions, appending each K/V once before attention. Caching saves recomputation of old rows — total attention work is still quadratic.

From loss to one update

Gradients, lifetimes, and one AdamW step

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The update creates a new model version: the old KV cache is invalid and must be rebuilt.

Toward Kimi K3

Interfaces kept, mechanisms changed

Interface	K3 mechanism
sequence mixing	Kimi Delta Attention, hybrid attention, Gated MLA
channel mixing	sparse MoE, Stable LatentMoE
depth mixing	Attention Residuals
input pathway	MoonViT and native multimodality

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Next: Chapter 2 maps the computational anatomy of K3 onto these same interfaces.